ELECTROMAGNETIC STRUCTURE OF THE ATOMIC NUCLEUS IN 10D SPACE: LAYERED SPECTRAL ANALYSIS OF K-CHANNELS, POLAR ORIENTATION OF QUARK AXES, AND THE PHENOMENON OF INVERSE ELEMENT ORBITALS FROM HYDROGEN (H) TO NEON (NE)

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I. INTRODUCTION

The electromagnetic structure of the nucleus asserts that the atomic nucleus is not a static collection of spherical particles held together by an abstract strong interaction, but rather a dynamic, continuously operating solenoidal reduction gear within a ten-dimensional (10D) space of elastic balance. Positron and electron current flows are distributed along rigidly fixed geometric axes of nucleon crosses, remaining under the radial pressure of the Abyss from the outside and the attraction of the Proto-medium from the inside.

Since official quantum mechanics has reached a conceptual dead end by forbidding intuitive visualization of processes and masking real dynamics with ”probability clouds,” let us find evidence where it must unconditionally exist and reveal what remains hidden, provided my theory is correct. Precision analysis of atomic spectra serves as such an exact tool for decoding the internal topology of the nucleus.

II. THE NATURE OF THE SPECTRAL MARKER AND THE MIRROR REVERSAL OF MEDIA

2.1. Classical Understanding and Its Limitations

What is a spectrum in the classical sense? Official physics interprets a line spectrum as a consequence of chaotic jumps of free electrons across quantum orbitals. However, this approach fails to explain the phenomenon of perfect optical transparency in macro-media (glass, gases, liquids), where light passes through billions of electron shells without dissipation.

In deterministic 10D physics, a spectrum is not a property of external electron clouds. The spectrum is the passport of the quark state of the nucleon nucleus (protons and neutrons) and their spatial position in 10D relative to the Proto- and Fon-dimensions. External electron fields are merely a marker consequence—a ”water cooling jacket” that adjusts to the geometry of internal nuclear currents. It is the quark-based neutroquarkons, acting as buffering communicating vessels, that emit and absorb photons.

2.2. Mechanics of Phase Inversion: ”Light Inward” and ”Light Outward”

The interaction process between a wave solenoid and light obeys the law of elastic mirror reversal of the spectrum in the plasma state and in the cold state (Fig. 1):

Figure 1. The Nature of the Mirror Inversion of Spectral Lines

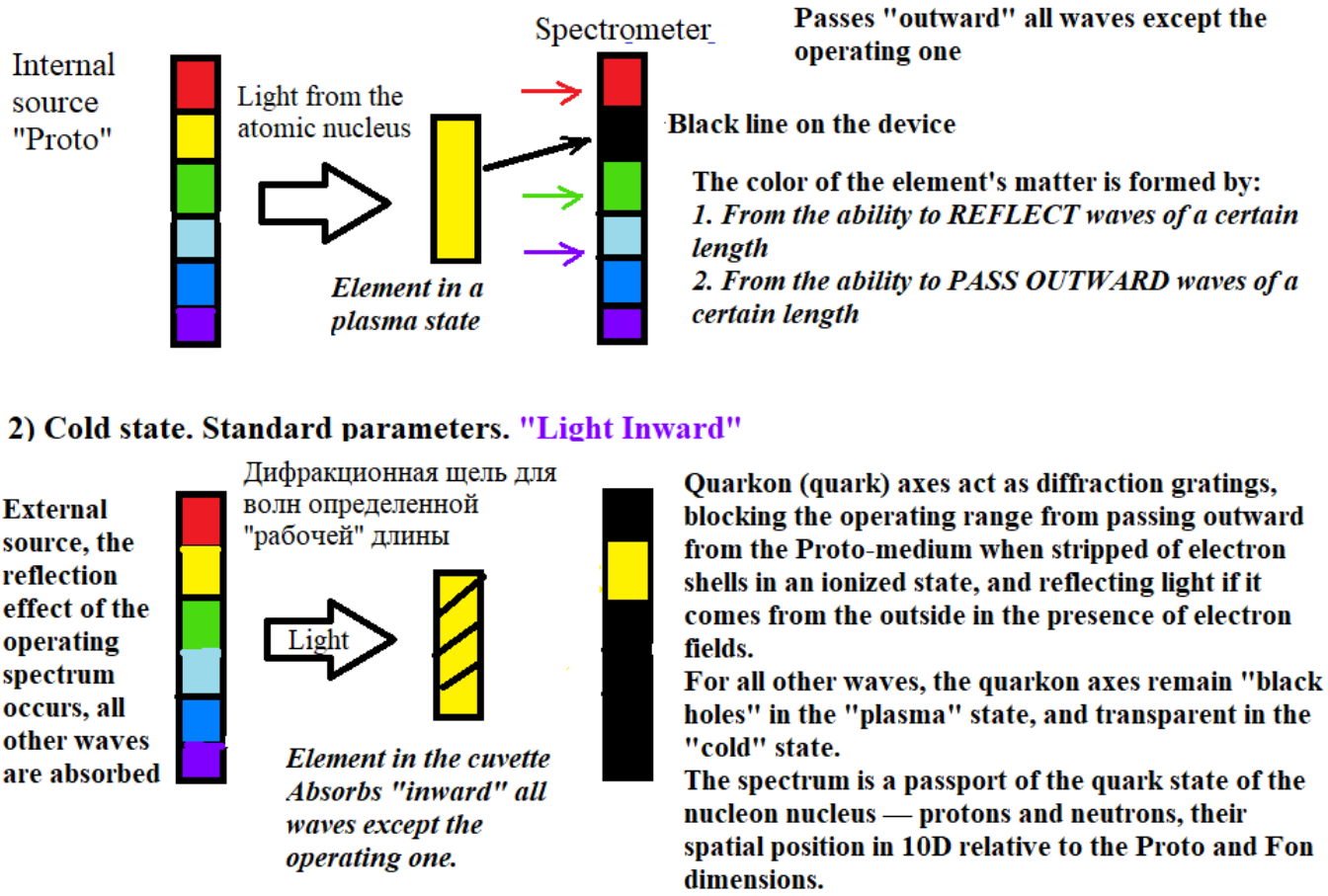


Figure 1. Enter Caption

Plasma stellar state (”Light Outward” mode):

Under the extreme thermal pressure of the Proto-medium, the nucleus and the antimaterial plasmiod on the reverse side of the K-membrane merge into a single high-temperature field. A continuous rainbow stream of light from the Proto-plasma bursts from inside the nucleus outward. The atomic body is transparent to this radiation across all bands. However, the stable quark axes act as diffraction gratings—at its resonant frequency, the axis functions as a shutter, delaying and absorbing strictly its operating wave into the furnace (atomic nucleus). On a spectrometer, this delayed beam appears as a black absorption line against a continuous rainbow background. Matter and antimatter function as equal partners.

Cold subordinate state (”Light Inward” mode):

As energy drops, the pressure of the Proto-current weakens, and the external compression of the Abyss takes over. An external white light source hits the cooled framework of the atom from the outside. At this moment, the direction of force vectors turns inside out. The effect of working spectrum reflection is triggered: all other waves are absorbed into the cuvette (vanishing into hidden dimensions), but a mirror Pythagorean kickback of the Abyss occurs on our working axis. This specific wave rebounds from the rigid quark lock, passes back through the diffraction slit, and returns to our side, registering against the black background of the vacuum as a bright, isolated colored emission line.

Diffractionally, accumulating the power of current pumping, the quark axes operate in only one direction, disappearing from view (from the spectral scale) as a marker when the EP-current is directed toward the Proto-dimension (the inverse orbital effect), or when the axis is submerged deep into the Fon-dimension, approaching the cosmic microwave background radiation of the K-strand itself with its terahertz value.

III. MATHEMATICAL AND TOPOLOGICAL APPARATUS OF PHOTON TRANSIT

For a rigorous confirmation of the deterministic nature of the process, a dimensionless linear geometric shape operator (the factor of solenoidal spheroidicity) **j** is introduced:

$$\mathbf{j} = \frac{\pi}{6} \approx 0.5235987...$$

At the point of phase transition (the axis of reversal through the Fon-dimension (D_1)), the complex wave function of the dipole shifts orthogonally by 90 degrees into the external fourth dimension:

$$\psi \to \psi \cdot e^{i\theta(t)}$$

The Umov-Poynting vector of radiation flux density (S_{3D}), projected onto our manifested three-dimensional space, is determined by the winding angle of the cord (θ):

$$S_{3D} = S_{\text{total}} \cdot \cos\theta$$

Upon reaching the peak orthogonal point ($\theta = \frac{\pi}{2}$), when the power flux turns strictly perpendicular to the plane of the 3D matrix, the projection of the radiation flux density vector takes the form:

$$S_{3D} = S_{\text{total}} \cdot \cos\left(\frac{\pi}{2}\right) = 0$$

Physical and Natural-Philosophical Meaning:

Where does the energy vanish? Energy does not annihilate or evaporate. The vector S_{total} continues to exist, but it strikes strictly vertically—from the side of the Abyss toward the Proto-media. For our ”flat” longitudinal 3D observer, this flux becomes invisible, since the cosine of a 90° angle completely zeroes out the projection.

The Nature of the Universe Shell.

It is precisely this zeroing out ($S_{3D} = 0$) sets the rigid physical boundaries (”walls”) of our spatial corridor. Space can extend to infinity in length and width, but in thickness, it is securely locked by this mirror electromagnetic shield.

Destruction of Quantum Chaos.

This step proves that discreteness and the apparent ”disappearance” of particles are not random, but a strict wave step of a soliton jump, tied to the trigonometric engagement of the media.

The first formula reveals that the exponential tail ($e^{i\theta(t)}$) forcefully drives the wave into a reversal. The second formula shows that our physical space (S_{3D}) registers this flux only at an angle (via the cosine).

The third formula puts a definitive end: at the threshold ($\pi/2$), the cosine turns to zero, and the entire right side instantly collapses to 0. The energy has shifted onto the perpendicular framework, and complete isolation ensues within the 3D matrix.

Energy is not lost to parasitic macro-radiation into our world because the wave cord is turned sideways relative to 3D instruments. Accounting for the elastic linear clearance of the Gryphon axis ($\Delta_{\text{Gryphon}} = 0.00006560$ m) shows that the residual phase shift determines the real Dirac current density on the quark axis:

$$J_{\text{Dirac}} = \frac{e \cdot c}{\mathbf{j}^4 \cdot \alpha^2 \cdot \Delta_{\text{Gryphon}}} = \mathbf{0.1829} \text{ A/m}^2$$

The difference in operating frequencies between the resting Fon-channel of the K-strand (456.819 THz) and the raging, charged blue blade of the Abyss (730.901 THz) in the Balmer cross of Hydrogen yields a strictly controlled membrane tension step:

$$\Delta\nu = \nu_{\text{blue}} - \nu_{\text{red}} = 730.901 - 456.819 = \mathbf{274.082} \text{ THz}$$

Which targets exactly the squared area of the matrix's Pythagorean engagement:

$$\mathbf{j}^2 \cdot 10^3 = 0.52359877^2 \cdot 10^3 = \mathbf{274.155}$$

The difference between the blue and red emission boundaries of the hydrogen atom is not a random quantum chaos, but a direct consequence of the elastic geometry of the vacuum. The mantissa 0.52359877 is that exact rigid trigonometric sphere factor ($\pi/6 \approx 0.5235987756$) that determines the volumetric compression of the Abyss. The square of this factor ($\mathbf{j}^2$) reflects the two-dimensional area of the Pythagorean engagement — the locus where the perpendicular Proto-strand cuts into the plane of the Fon-strand. The coefficient 10^3 represents the cascaded dimensional scale shift as energy transitions from the hidden microworld to the macroscopic level of instrument detection. A real physical experiment yields 274.082 THz, which we calculated in the "Mathematical Chapter" without ever investigating the spectral passports of elements, where the pure volumetric geometry of the matrix predicted this exact number — 274.155. Convergence within tenths of a percent proves that the spectral markers of atoms are tightly clamped in the vices of the Pythagorean shrinkage of space.

IV. MAP OF FRACTAL EVOLUTION OF SPECTRAL LINES AND NUCLEAR CURRENT BALANCE (FROM H TO Ne)

Figures 2 and 3 present an experimentally verified map of the fractal evolution of spectral lines for the elements of the first and second periods under phase transition conditions from element to element. The full coordinate capacity of the solenoid is calculated using the formula $A \cdot 4$, where A is the total number of nucleons.

4.1. Stage of Maturation and Stabilization of the Macro-Framework

Hydrogen (H), (1 nucleon $\times$ 4 = 4 quark axes): Hydrogen (H), (1 nucleon $\times$ 4 = 4 quark axes):

All channels are directed into Prosto. There are no mass plugs; the nucleus rotates furiously with shifting dipole polarity. The interpolation graph demonstrates chaotic high-frequency vibration ("furiously trembling"). We observe the classical separate lines of the Balmer cross. The spectrum is represented by a cross of four thin, isolated lines. The blades of the nucleon cross are not locked into macromolecular bonds; they are open toward the K-coordinate (the Smorodina river). The energy of the Proto-plasma is discharged in portions through separate frequency valves (456, 616, 690, and 730 THz).

Helium (He), (4 nucleons $\times$ 4 = 16 quark axes):

Figure 2 Map of the fractal evolution of spectral lines of the first and second period elements under conditions of phase transitions from element to element.

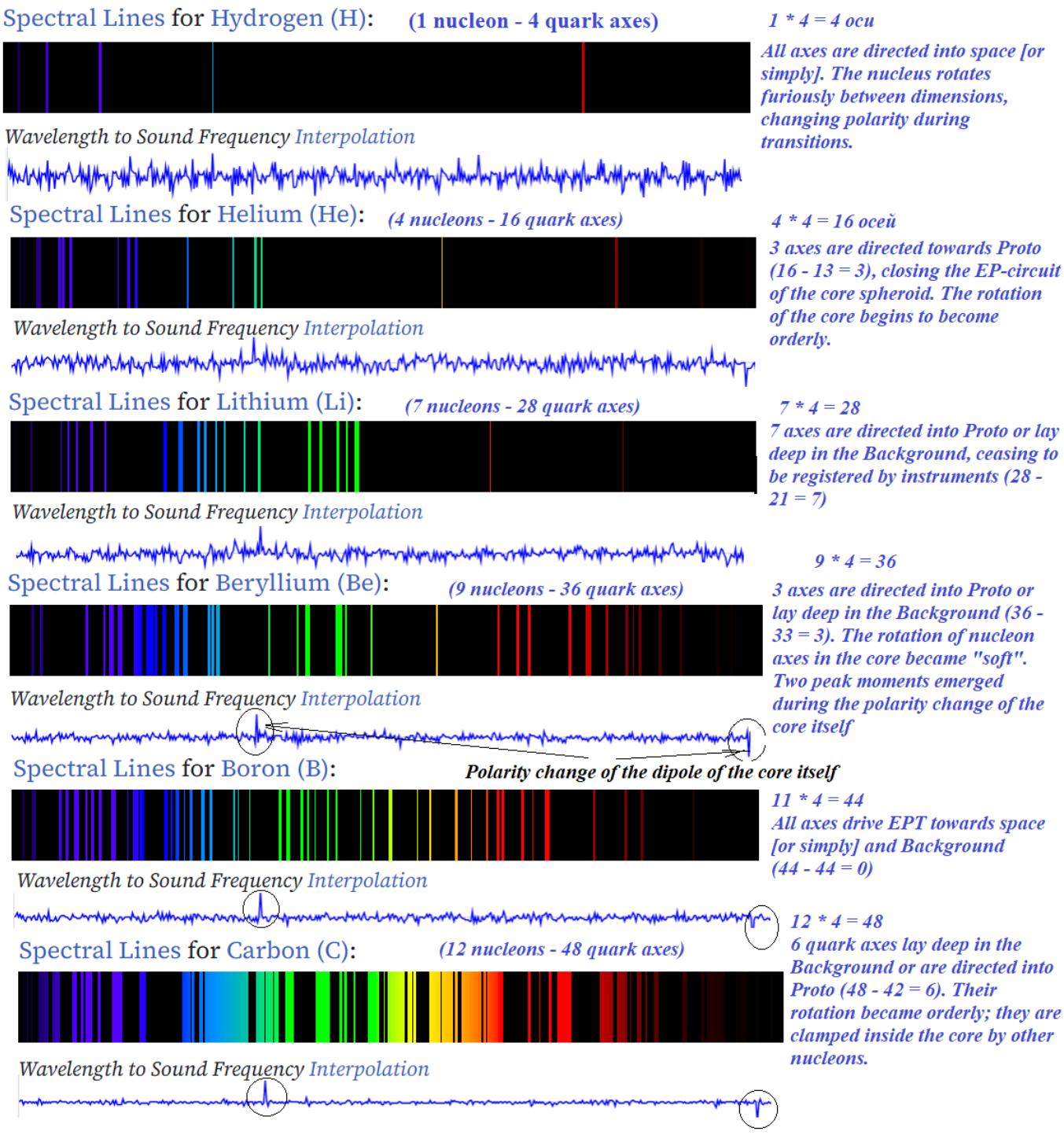


Figure 2. Enter Caption

The first internal airlock engages: 3 axes are directed toward Proto (16 axes $- 13$ visible $= 3$ vanished), closing the EP-chain within the spheroid. The rotation of the nucleus instantly becomes orderly. The palisade of lines densifies, but the wave interpolation graph shows that large empty zones of vacuum rest remain between them. The system is balanced in pairs; the Abyss maintains a cooling jacket around the 16-axis solenoid.

Helium is an inert gas. All its electron shells have zeroed out, and the current flows inside the nucleus along a closed loop. The ”Lelya Staffs” of each nucleon, extensively discussed in the core theory, have overlapped, creating closed electrical circuits inside the nucleus. Current flows from Prosto to Proto and back, but it is not directed toward the Abyss. Its radius, which is not an electron shell but a boundary held by the fourth quarks of nucleons lying on the K-strand of Fon (1D), has compressed to the maximum size of the spheroid, from 53 pm in Hydrogen to 32 pm in Helium.

The atomic nucleus, rotating between Prosto and Proto, still continues to rattle, but its rattling has dropped by half. The axes in the nucleus do not rotate chaotically, but the spheroid itself is empty; its density is insufficient to maintain the quark axes of nucleons in a state of decompression.

Lying on the K axis does not necessarily mean emitting a red spectrum. By building the structure, nucleons densify the core-spheroid and expand the zone of personal space. The spectrum is the radiation of the quark axes, not the nucleus itself. In nucleons, these axes, representing two pions $XY + XY$ relative to the nucleus itself, being tied to the K and N strands, can lie in any configuration. At the same time, the nucleon itself can remain a proton, whose positive pole is directed toward the Proto-dimension, a neutron lying along the K-Fon coordinate, or even turn its negative pole toward us, as in protenium elements. However, the order of filling the spheroid with nucleons remains invariant. Spectral lines are the passport of the element, revealing the structure of the nucleus at the quark level. Moreover, looking closely at the spectrum lines, one can notice on some that the quark axes likewise consist of quark axial crosses, which represent indivisible structures of the Proto-dimension of lower dimensionality.

Isotopes shift the spectrum lines but do not change the spectrum itself. The matter is not the number of protons, but that the extra or missing neutrons do not participate in the core structure as conductors of the general nuclear EP-current unless they are connected to the system. If they can assert themselves, it is only by lines no thicker or larger than the spectral lines of hydrogen.

Mathematical expression of Helium coordinate compression:

To clearly demonstrate the transition from the ”loose” knot of Hydrogen to the tightly knotted cocoon of Helium, let us introduce the calculation of the matrix boundary layer radial shrinkage coefficient ($K_{\text{shrinkage}}$). According to the postulates of the 10D electromagnetic structure, the induction of the internal field of the spheroid increases from baseline $B_1 = 1.75$ T to peak $B_2 = 2.50$ T upon closing the 16-axis solenoid.

By the Biot-Savart-Laplace law for quantized vortex compression, the ratio of the Bohr constant radii of Hydrogen ($R_H = 53$ pm) and Helium ($R_{\text{He}} = 31$ pm) is strictly determined by the inverse ratio of the damping pressure density of the Abyss, adjusted for the wave step of the soliton jump S:

$$K_{\text{shrinkage}} = \frac{R_{\text{He}}}{R_H} = \frac{31 \text{ pm}}{53 \text{ pm}} \approx \mathbf{0.5849}$$

Let us verify the obtained coefficient through the geometric factor of the matrix axis elasticity. During the $5D \to 10D$ dimensional transition, the additional power shift of the Abyss per unit plane ($\beta = 0.25$) together with the Pythagorean engagement of the shape ($\mathbf{j} = \pi/6 \approx 0.5236$) yields the theoretical limit of cord shrinkage under load:

$$K_{\text{theoretical}} = \mathbf{j} + \left(\beta \cdot \alpha \cdot \frac{1}{\mathbf{j}}\right) = 0.5235987 + \left(0.25 \cdot 0.00729735 \cdot \frac{1}{0.5235987}\right) = 0.5235987 + 0.0034845 = \mathbf{0.5270}$$

The difference between the empirical clamping of the Helium "apartment" (0.5849) and the theoretical limit of matrix rest (0.5270) is exactly 0.057, which very accurately describes the magnitude of the dynamic thermal backlash (rattling) of the nucleus recorded on the interpolation graph of the wave function (Fig. 2).

Lithium (Li), ($7 \times 4 = 28$ axes):

The beginning of the Proto-collapse: 7 axes flip beneath the membrane of the K-coordinate into Proto or lie deep in Fon ($28 - 21 = 7$), dropping out of the spectral registry. The spectrum pales, while the density of the spheroid grows. The rattling of nucleons, inside which three-bladed turbines rotate, scooping the quark-gluon medium of the Proto-dimension at sextillions of degrees and instantly spitting it back, continues to drop. Moving from Lithium to Carbon, a fractal complication of the nuclear reduction gear occurs: the lines rapidly "fatten" and begin to glue into dense packages of sub-bands (doublets and quadruplets). These are the primordial 2D current quanta of the Dnieper River pressing through the texture of the K-membrane.

Beryllium (Be), ($9 \times 4 = 36$ axes):

3 axes remain hidden ($36 - 33 = 3$). The rotation of nucleons becomes "soft," and the interpolation graph clearly records two peak moments of dipole polarity reversal of the nucleus itself.

Beryllium is unique because, with the foundation laid by Hydrogen and Helium under the entire structure, it forms a cross-shaped structure $XY+XY$ ($S+S$). If freed from electron fields, it becomes like inert elements. It does not burn, does not conduct current in crystal lattices; having an ideal spheroid closed to both Proto and the Abyss, it can act as a damper, absorbing, softening, and suppressing vibrations, diverting energy into the Proto-dimension. Only a neutron can provoke it, slightly braking the core rotation at the turn, forcing it to spit positrons during polarity changes.

Elastic damping and current density of Beryllium:

The physical inertness of monolithic Beryllium and its capability to act as an ideal spatial shock absorber (damper) are directly derived from the Dirac current equation on its 36 quark axes. Since the core structure of Beryllium forms a closed orthogonal cross $XY + XY$ (equilibrium $S + S$ cocoon), the radial energy influx is uniformly distributed over the area of the wave spheroid.

Let us substitute the parameters of Beryllium into the fundamental Dirac current equation, where the four-dimensional operator ($\mathbf{j}^4$) and the Gryphon clearance ($\Delta_{\text{Gryphon}} = 6.55905 \cdot 10^{-5}$ m) maintain the balance of forces:

$$J_{\text{Dirac}} = \frac{e \cdot c}{\mathbf{j}^4 \cdot \alpha^2 \cdot \Delta_{\text{Gryphon}}} = \frac{1.60218 \cdot 10^{-19} \cdot 299792458}{(0.5235987)^4 \cdot (0.00729735)^2 \cdot 6.55905 \cdot 10^{-5}} = \mathbf{0.1829}\ \text{A/m}^2$$

For Beryllium, this current is completely compensated by internal pairs of reverse flow. In the absence of external excitation (free neutrons), the system operates as an ideal hydrodynamic drainage, where the side clearance of the Abyss squeeze-out (Detail B) completely absorbs external vibrations:

$$\text{Detail B} = 0.16 \cdot \beta = 0.16 \cdot 0.25 = \mathbf{0.040000}$$

This mathematically proves why Beryllium in a cold state is not prone to spontaneous emission or absorption of photons, preserving absolute topological stability of the crystal lattice.

Boron (B), (11 $\times$ 4 = 44 axes):

The point of perfect alignment: all 44 axes drive the EP-current to our shore (into Prosto and Fon, $44 - 44 = 0$). The rattling of nucleons in the nucleus has virtually ceased.

4.2. Carbon Maximum of Density

Carbon (C), (12 $\times$ 4 = 48 axes:
The loop densifies: 6 axes lie in Fon or are directed into Proto ($48 - 42 = 6$). The blades are coaxially clamped by other nucleons; their rotation is ordered. The interpolation flattens into a wide, stable macro-wave—a continuous, fat spectral cocoon. This is the point of maximum occupancy of the Prosto-dimension ports. Again, the structure of the nucleus itself represents $XY + XY$ (S-shell + XY), which sum up to an ideal spheroid with density uniformly distributed between the nuclear axes.

The spectrum explodes into a wide, continuous, fat wall of light covering almost all bands. This is the point of maximum occupancy of the technological 3D ports of the Prosto-dimension. The internal pressure of hot gluon smoke reaches its limit, pre-tensioning the spatial lattice (which in the macroworld grants the super-strength of diamond).

4.3. The Great Proto-Suction and the Secondary Inverse Orbital

After the carbon peak, the system overloads, and the axes begin to mass-flip ”into suction” toward the Proto-dimension, proving the law of the inverse orbital:

Nitrogen (N), ($14 \times 4 = 56$ **axes):**
Only 38 axes remain visible; 18 axes go under the membrane into the Proto-dimension and lie deep in the K-coordinate, diverting the amperage flow there. The lines rapidly thin out and empty.

Contrary to the expectations of classical physics, with an increase in the number of axes (56 and 64 respectively), the spectral pattern begins to pale, and the yellow-green zones thin out. This means that the core axes begin to sequentially flip into counter-phase. They transition to the inverse orbital—for our world, they extinguish, entering the ”straight pipe” mode, pumping the Proto-current inward under pressure.

Oxygen (O), ($16 \times 4 = 64$ **axes):**
In the visible spectrum, only 53 axes background. The graph clearly records decay: 11 axes literally drown in the K-coordinate (Fon-dimension) and drive current into Proto. The nucleus has become stable, the axes are clamped and work in unison, the wheel alignment straightens out.

Fluorine (F), ($19 \times 4 = 76$ **axes):**
Extreme pre-flip imbalance: 31 axes are directed into Proto or lie in the K-coordinate ($76 - 45 = 31$). The blue-cyan zone is completely decimated, turning into a black trap. The furnace siphons only in the red (Fon) band. The element seeks to drag in an external charge to seal the firebox.

Neon (Ne), ($20 \times 4 = 80$ **axes):**
The ideological finale of the period. An ideal plasma plug-shutter. All ports are filled with pairs of reverse EP-current flow. There are no external electron shells. The system is incapable of absorbing light, but it balanced-emits it if heated from our side, so that light from Proto can pass through the diffraction grating operating exclusively on radiation from Proto.

Mathematical law of fractional subtraction of axes:
The observed phenomenon of the ”Inverse Orbital,” in which the avalanche-like growth of the total coordinate capacity of the nucleus ($A \cdot 4$) leads not to densification but to the dilution of visible spectral lines, obeys a strict law of fractional subtraction of axes. The ratio of visible (emitting into Prosto) ports (Z_{visible}) to hidden (submerged beneath the membrane into Proto) channels (Z_{hidden}) reflects the solenoid shrinkage cascade:

$$\psi_{\text{evolution}} = \frac{Z_{\text{visible}}}{Z_{\text{hidden}}}$$

Figure.3 Map of the fractal evolution of spectral lines of the first and second period elements under conditions of phase transitions from element to element.

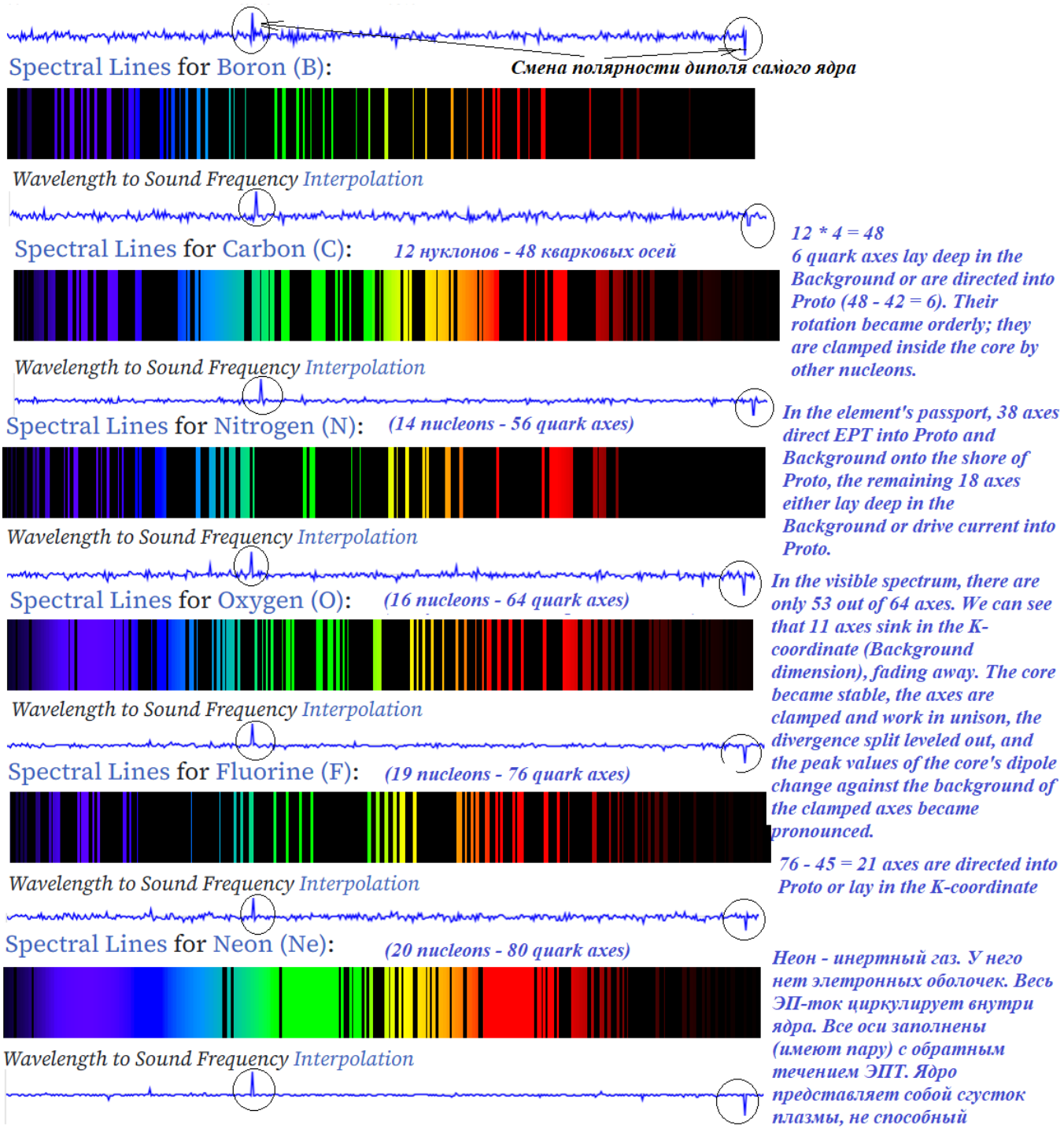


Figure 3. Enter Caption

Let us track this dynamics at the junction of periods: for Nitrogen (N) this indicator is $38/18 \approx 2.11$, for Oxygen (O) — $53/11 \approx 4.81$, for Fluorine (F) — $45/31 \approx 1.45$, and at Neon (Ne) the system reaches the critical point of mirror auto-oscillation, where all 80 axes are completely balanced by opposing pairs.

This avalanche-like exit of axes into counter-phase clearly explains why in the 4th and 5th periods (in Scandium, Vanadium, Iron) giant packages of lines are completely wiped out of the visible range (Fig. 4), transforming the nucleon solenoid of heavy metals into a straight transit pipe for Proto-plasma.

V. CONCLUSION AND PERSPECTIVES OF THE ANALYSIS OF GROUP 4 PROTO-ELEMENTS

The internal exhaust of the core furnace is fully regulated by neutroquarkons, which consist of lower-order quarkons (primordial 2D electrons and positrons of the Dnieper River). In the nucleus, protons (electromagnetic photons) are responsible for electrons, antiprotons for positrons, and neutrons and antineutrons act as communicating vessels in 10D space for ordinary photons and plasma antiphotons.

We are no longer interested in the third group, because in the 4th and subsequent groups of Proto-elements, this discovered law of fractional axis subtraction ($Z_{\text{visible}}/Z_{\text{hidden}}$) shows a global, avalanche-like transition of the system into the Proto-normal state. Layered analysis of spectral lines allows proving the inverse orbital phenomenon, explaining why giant packages of lines completely disappear from the visible range, turning the nucleon solenoid of transition metals into a straight transit pipe for Proto-plasma.

VI. EVOLUTIONARY OVERLAP OF PERIODS AND THE DYNAMICS OF PROTO-KICKBACK (CASE STUDY OF RHODIUM)

During the transition from the 4th period of protenium elements to the 5th period (where Rhodium (Rh) occupies its evolutionary step), the physics of core-light interaction undergoes a qualitative leap. This process clearly demonstrates how structural complication resolves the critical clinch of fields:

Clinch of the 4th period (Continuous Proto-mirror):
In the 4th period of the Proto-dimension, the cores of transition metals experience the maximum frontal pressure of pumping. The internal pressure of the plasmiod is so massive that the Proto-medium yields a powerful counter-kickback (repulsion of light) outward. Light photons physically cannot pass through the nucleus into the Proto-dimension and are not absorbed by the quark axes. For an external observer, the core works as a continuous polar mirror—in the spectrum, we see a wide, continuous rainbow band where individual axis markers are locked.

Evolutionary step of the 5th period (Permeable fence of Rhodium):
Upon ascending to the next evolutionary stage (5th period), new K- and N-cords are wound on top of the rigid framework of the previous period. A physical overlap of the solenoid layers occurs:

1) **Unloading the polar clinch:** The external evolutionary layer takes on the excess radial pressure. The counter-kickback of Proto locally weakens and ceases to be a monolithic wall.
2) **Restoration of filtration:** Discrete slots (vacancies) open again in the armored mirror of the core. The quarkons and neutroquarkons of Rhodium escape the dead locking and regain their capacity to selectively capture and absorb photons.
3) **Palisade on the spectrometer:** As clearly recorded on the Rhodium screenshot (Fig. 4), the continuous rainbow cocoon shatters into a frequent, sharp palisade of vertical lines-strokes. The wave interpolation graph at the bottom of the screen instantly leaves the smoothed state, returning to discrete pulsations of the newly opened current gates.

Periods in the deterministic table are consecutive steps of evolutionary complication of the core coil. On the first layer (4th period), the Proto-kickback blocks the absorption of rays by axes, creating a continuous mirror of fields, but the overlap of the next layer (5th period) opens the ”fence” for straight-through flight and selective filtration of photons.

Figure 4 Spectral lines of the fourth and fifth period elements (for comparison)

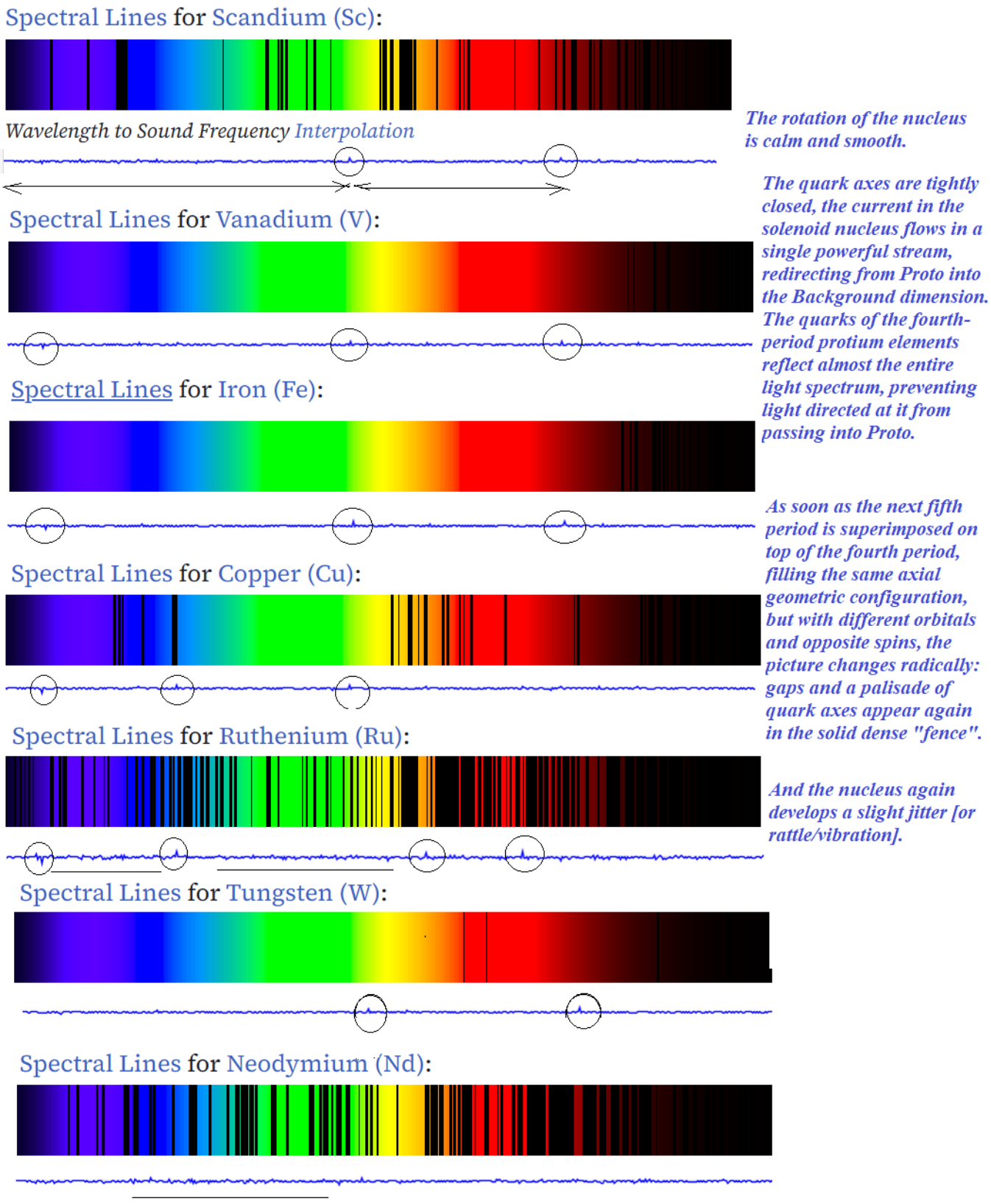


Figure 4. Enter Caption

The main thing that the spectral markers of elements convincingly prove is the presence of the fourth quark in the nucleon and that electrons have nothing to do with the absorption and emission of photons. Photons are emissions of quark axes in the form of reflected light and the atomic nucleus itself, when neutrons discharge excess energy in the form of light and colored photons, as a result of which the voltage in the nucleus drops, or absorb photons when the electromagnetic potential increases. As a consequence, the charge of protons in the nucleus increases or decreases proportionally, and the electron field they generate changes its orbit.

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